BIG MODULAR | PD-ONLY 9 V | ONE CONNECTED CIRCUIT | A1 WIP

STOP: PCB has only J1 and mechanical holes. No tracks / vias / zones. Neither board is operational.

5 V VBUS powers negotiation only. Gate = PG active LOW AND 7.88..10.78 V nominal window AND RUN. NO 5 V OUTPUT MODE.

BLOCKERS: PG does NOT prove PDO/current. No qualified input ESD/surge protection. J1 has four physical DRC violations.

9 V must be independently verified by PD analyzer at >=0.5 A contract before any test. About 40 mA/rail simultaneous TEST LOADS ONLY: no rating.

USB-C / FUSE

eFUSE / UV + OV

4.016 V BUCK

BIPOLAR +/-13.95 V

+12 V POSTREG

-12 V POSTREG

OFF-BOARD J2

9 V REQUEST / PG INTERLOCK

6.52 V BUCK

+5 V POSTREG

R4/R5: 562k/100k UV; R6/R7: 806k/100k OV.
SHDN LOW = OFF. JP1 cannot override either window limit.
PG generic active-low OD, not authenticated contract/current.
Default RUN OPEN; RTN is NOT externally shorted to GND.

U4 PSP/PSN grounded: BSW intentionally unused.
Positive boost can preBias only AFTER the gated V4 supply exists.
Negative feedback lower resistor returns to BI_REF, not GND.
PG of U3 holds both U4 enables low during intermediate startup.

POWER/RETURN DECLARATIONS (not PCB pads)

No TVS fitted: D1 reference reserved. Do not substitute a 12 V-standoff clamp on upstream USB VBUS.
LDOs are not precision output-OVP devices. No all-rail shutdown/tracking or rapid discharge guarantee.
Read docs/design-review.md, mechanics.md and reports/validation-summary.md. Converter stability, thermals, surge and physical fit remain untested.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

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